

Ryan Quach

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EDUCATION

University of California, Los Angeles

Bachelor of Science (B.S.) in Statistics

Masters of Science (M.S.) in Statistics

Honors: Phi Beta Kappa, Dean's Honor List | GPA: 3.92/4.0 | June 2025

GPA: 3.96/4.0 | June 2027

EXPERIENCE

Southern California Edison

Data Science Intern, System Planning & Engineering

Pomona, CA

June 2026 - Present

- Build statistical models to predict probability of failure (POF) for electric equipment, including underground switches
- Create deep learning models to determine outage cause from troublemen's text comments based on BERT embeddings
- Query and wrangle millions of rows of customer data related to outages and repair orders to identify areas for improvement

UCLA Department of Statistics and Data Science

Teaching Assistant

Los Angeles, CA

March 2026 - Present

- Lead discussion sections to review homework questions and elaborate on concepts discussed in lectures
- Deliver actionable feedback by grading homework assignments and midterm examinations
- Proctor midterm exams directly to ensure an orderly test-taking environment for the students

UCLA Herbarium, Mathias Botanical Garden

Research Intern

Los Angeles, CA

October 2024 - June 2025

- Facilitated the physical refileing and digital imaging of botanical specimens within the university collection
- Investigated the most common families, genera, and other categories for plants in the Herbarium using R
- Visualized the distribution of botanical field findings within California through R geospatial packages

PROJECTS

Predicting Wildfire Severity in Northeastern Portugal | *tidyverse, tidybayes, brms, sf*

April 2026 - June 2026

- Created a Bayesian hierarchical model to estimate wildfire size within Montesinho National Park in NE Portugal
- Calculated posterior distributions for impacts of meteorological variables such as wind speed and relative humidity
- Established predictions for wildfire severity on an entire spatial grid mapping out the entirety of the national park

Causal Inference with Health Insurance Data | *tidyverse, gt, sensemakr, data.table*

January 2026 - March 2026

- Estimated causal impact of health insurance type on hospital mortality and elopement rates
- Used data from MIMIC-IV, which was collected at the Beth Israel Deaconess Medical Center
- Employed augmented inverse propensity weighting (AIPW) based on linear regression to calculate causal effects

Modeling Food Delivery Times | *Scikit-learn, XGBoost, Polars, Seaborn, Matplotlib*

November 2023 - January 2024

- Utilized food delivery data from the Indian subcontinent to estimate food delivery times
- Incorporated several machine learning algorithms such as linear regression and gradient boosting
- Determined that average prediction error was less than one-third of the standard deviation of the delivery times

Predicting Customer Churn | *Scikit-learn, XGBoost, Polars, Seaborn, Matplotlib*

February 2023 - April 2023

- Extracted dataset from anonymous telecommunications firm and tidied data with the polars package
- Forecasted the rate of customer churn with the assistance of scikit-learn and XGBoost
- Calculated business value of about \$1.2 million annually for every ten thousand customers

SKILLS

Data Analysis / Data Wrangling: tidyverse, pandas, polars

Data Visualization: Canva, ggplot2, Plotly, Seaborn, Matplotlib, Shiny, Streamlit

Machine Learning / Statistical Learning: Caret, scikit-learn, XGBoost, brms

Microsoft Applications: Excel, Powerpoint, MS Teams

Programming Languages: Python, R, SQL